



Sudoku in Graph Theory

Taiki Aiba

Mentor: Tantan Dai

Spring 2024 Directed Reading Program

April 18, 2024

Overview

- What is Sudoku?
- Sudoku and Graphs
- Extendable Colorings
- Sudoku Number
- Some Results

What is Sudoku?

Before we dive into some of the combinatorial aspects of Sudoku, we shall first briefly explain how Sudoku works.

Rules

A Sudoku puzzle typically consists of a 9×9 square grid divided into nine 3×3 smaller grids. Each square must be filled in with a number from 1 to 9 such that each row, column, and 3×3 grid contains every number from 1 to 9 exactly once. At the start, some squares will already be filled in.

What is Sudoku?

Example

	7	5		9				6	1	7	5	2	9	4	8	3	6
	2	3		8				4	6	2	3	1	8	7	9	4	5
8						3			8	9	4	5	6	3	2	7	1
5			7		2				5	1	9	7	3	2	4	6	8
	4		8		6		2		3	4	7	8	5	6	1	2	9
			9		1			3	2	8	6	9	4	1	7	5	3
9			4					7	9	3	8	4	2	5	6	1	7
	6			7		5	8		4	6	1	3	7	9	5	8	2
7				1		3	9		7	5	2	6	1	8	3	9	4

What is Sudoku?

So now we ask a question:

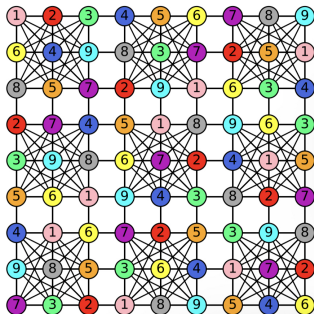
Question

When does a Sudoku board have a *unique* solution, and how can we characterize this condition?

Sudoku and Graphs

We will use graph theory!

1						7		9
	4				7	2		
8								
	7			1			6	
3								5
	6			4			2	
								8
		5	3				7	
7		2					4	6



Terms

A graph $G = (V, E)$ is a structure consisting of vertices from a set V and edges from a set E such that each edge connects two distinct vertices.

A k -coloring of a graph G is a coloring $c : V \rightarrow [k]$ of the vertices of G such that every edge has differently-colored endpoints.

The chromatic number $\chi(G)$ of a graph G is the smallest k such that G has a k -coloring.

We can view a Sudoku puzzle as a graph $G = (V, E)$, where $\chi(G) = 9$. The solution to a Sudoku corresponds to a 9-coloring of G , and solving a Sudoku corresponds to “extending” a partial coloring of G , where this extension is unique.

Extendable Colorings

Let's go over some definitions related to the notion of “extending” a Sudoku board.

Definitions

Let $G = (V, E)$ be a connected graph with $|V| = n$. Suppose $\chi(G) = k \geq 2$. Let $S \subseteq V$, and let c be a proper coloring of $G[S]$ (the graph with vertex set S and all edges that connect two vertices in S).

c is an **extendable coloring** if c can be extended into a k -coloring of G ; i.e., if there exists a solution to the Sudoku board.

c is a **uniquely extendable coloring** if c can be uniquely extended into a k -coloring of G ; i.e., if there exists a unique solution to the Sudoku board.

Definitions

Let $G = (V, E)$ be a graph with $\chi(G) = k$.

A set of vertices $S \subseteq V$ with a coloring $c : S \rightarrow [k]$ is said to be a **defining set** of G if there exists a unique extension of the colors of S into a k -coloring of G .

A defining set of minimum cardinality is a **minimum defining set**.

Traditionally, the size of a minimum defining set of a graph G is denoted by $d_V(G)$. In other literature, it is also known as the **sudoku number** of G , denoted by $sn(G)$.

Some Results

It was proven that the sudoku number of a 9×9 Sudoku board is **17**. In other words, if we have less than 17 givens, then we cannot possibly have a unique solution. One paper that proved this claim used a technique called “hitting set enumeration”.

It was also proven that we need at least 8 *distinct* givens (i.e., at least one given that is each of eight of the nine possible digits) for there to possibly exist a unique solution. In general, if we have an $n \times n$ Sudoku board, then we need at least $n - 1$ distinct givens.

Some Results

It is possible that even if we have as many as 77 givens, we could still not have a unique solution. Below is an example.

9	2	6	5	7	1	4	8	3
3	5	1	4	8	6	2	7	9
8	7	4	9	2	3	5	1	6
5	8	2	3	6	7	1	9	4
1	4	9	2	5	8	3	6	7
7	6	3	1			8	2	5
2	3	8	7			6	5	1
6	1	7	8	3	5	9	4	2
4	9	5	6	1	2	7	3	8

9	4
4	9

4	9
9	4

Conclusion

Sudoku offers a wealth of fun and intriguing math insights. As more researchers have looked into Sudoku from a graph theoretic perspective, more information and results have been discovered about the nature of this iconic puzzle.

References

-  Gee-Choon Lau, J. Maria Jeyaseeli, Wai-Chee Shiu, and S. Arumugam.
Sudoku number of graphs, 2022.
-  Gary McGuire, Bastian Tugemann, and Gilles Civario.
There is no 16-clue sudoku: Solving the sudoku minimum number of clues problem via hitting set enumeration.
Experimental Mathematics, 23(2):190–217, 2014.
-  Alexey Pokrovskiy.
Graphs with sudoku number $n - 1$, 2022.